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This is the errata for the book

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The author's copy with errors corrected can be found in the following link:

`https://xiaoluhou.github.io/Textbook.pdf`

Location	Original	Change
Page 8, Theorem 1.1.4 (Euclid's division)	Given $m, n \in \mathbb{Z}$, take q, r such that $n = qm + r$. Then $\gcd(m, n) = \gcd(m, r)$.	Given $m, n \in \mathbb{Z}$, take $q, r \in \mathbb{Z}$ such that $n = qm + r$, and $0 \le r < n$. Then $\gcd(m, n) = \gcd(m, r)$.
Page 16, Definition 1.2.8	The distributive laws holds	The distributive laws hold
Page 20, Example 1.2.24	$f(1 \oplus 0) = f(1) = a,\ f(1) + f(0) = a + b = a$	$f(1 \oplus 0) = f(1) = b,\ f(1) + f(0) = b + a = b$
Page 9, Algorithm 1.1, lines 2-4	**Input:** $m,\ n$// $m, n \in \mathbb{Z},\ m \neq 0$ **Output:** $\gcd(m, n)$ 1 **while** $m \neq 0$ **do** 2 $r = n\%m$// remainder of n divided by m 3 $n = m$ 4 $m = r$ 5 **return** r	**Input:** $m,\ n$// $m, n \in \mathbb{Z},\ m \neq 0$ **Output:** $\gcd(m, n)$ 1 **while** $m \neq 0$ **do** 2 $r = m$ 3 $m = n\%m$// remainder of n divided by m 4 $n = r$ 5 **return** n
Page 18, first paragraph below Definition 1.2.12	By definition, for any $a \in F$, there exists $b \in F$ such that ...	By definition, for any $a \in F$, $a \neq 0$, there exists $b \in F$ such that ...
Page 20, Example 1.2.24	$f(1 \oplus 0) = f(1) = a,\ f(1) + f(0) = a + b = a$	$f(1 \oplus 0) = f(1) = b,\ f(1) + f(0) = b + a = b$
Page 21, last paragraph	If $a_{ij} = 0$ for $i \neq j$, A is said to be a *diagonal matrix*. An $n-$*dimensional identity matrix*, denoted I_n, is a diagonal matrix whose diagonal entries are 1, i.e. $a_{ii} = 1$ for $i = 0, 1, \ldots, n-1$. ... An $n \times n$ matrix is called a *square matrix* (i.e. a matrix with the same number of rows and columns).	An $n \times n$ matrix is called a *square matrix* (i.e. a matrix with the same number of rows and columns). If A is a square matrix and $a_{ij} = 0$ for $i \neq j$, then A is said to be a *diagonal matrix*. An $n \times n$ *identity matrix*, denoted I_n, is an $n \times n$ diagonal matrix whose diagonal entries are 1 and all the other entries are 0, i.e. $a_{ii} = 1$ for $i = 0, 1, \ldots, n-1$ and $a_{ij} = 0$ for $i \neq j$. ...
Page 44, above Example 1.4.17	$x = \sum_{i=1}^{3} a_i y_i M_i \bmod n = 2 \times 2 \times 35 + \ldots$	$x = \sum_{i=1}^{3} a_i y_i M_i \bmod m = 2 \times 2 \times 35 + \ldots$
Page 44, Example 1.4.17	With the extended Euclidean algorithm, we have found	With the extended Euclidean algorithm, we can find
Page 48, Equation (1.23)	$f(x) \times_{F[x]} g(x) := d_n x^n + d_{n-1} x^{n-1} + \ldots$	$f(x) \times_{F[x]} g(x) := d_{m+n} x^{m+n} + d_{m+n-1} x^{m+n-1} + \ldots$
Page 49, Theorem 1.5.1	of $\deg(f(x)) \ge 1$	if $\deg(f(x)) \ge 1$
Page 51, Example 1.5.6	$\mathbb{F}_2[x]/(f(x)) = \{1, x, x+1\}$ ... $\mathbb{F}_2[x]/(g(x)) = \{1, x, x+1\}$	$\mathbb{F}_2[x]/(f(x)) = \{0, 1, x, x+1\}$ ... $\mathbb{F}_2[x]/(g(x)) = \{0, 1, x, x+1\}$
Page 59, Definition 1.6.6	A binary code C is said to be $k-$*error correcting* if the minimum distance decoding outputs the correct codeword...	A binary code C is said to be $k-$*error correcting* if with the incomplete decoding rule, minimum distance decoding outputs the correct codeword...
Page 72, Example 1.7.11	$1_A : \mathcal{A} \to \mathbb{R}, \quad 1_A(\omega) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}.$	$1_A : \Omega \to \mathbb{R}, \quad 1_A(\omega) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}.$
Page 81, below Definition 1.7.11	For example, if when X is large (or small), Y is also large (or small), then the signs of $X_i - \overline{X}$ and $Y_i - \overline{Y}$ will tend to be the same. Or if when X is large (or small), Y is small (or large), then the signs of $X_i - \overline{X}$ and $Y_i - \overline{Y}$ will tend to be different.	For example, if when X is large (or small), Y is also large (or small), then the signs of $X_i - \mu_X$ and $Y_i - \mu_Y$ will tend to be the same. Or if when X is large (or small), Y is small (or large), then the signs of $X_i - \mu_X$ and $Y_i - \mu_Y$ will tend to be different.
Page 105	$\boldsymbol{c} = c_1 c_2 \cdots = e_k(p_1) e_k(p_2) \ldots.$	$\boldsymbol{c} = c_1 c_2 \cdots = E_k(p_1) E_k(p_2) \ldots.$
Page 105	$\boldsymbol{c} = c_1 c_2 \cdots = e_{z_1}(p_1) e_{z_2}(p_2) \ldots$	$\boldsymbol{c} = E_{z_1}(p_1) E_{z_2}(p_2) \ldots$
Page 106 Table 2.2 (b)	Á 11000001 C1 Ä 11000100 C4 Í 11001101 CD × 11010111 D7 ÷ 11110111 F7	Á 1100001110000001 C381 Ä 1100001110000100 C384 Í 1100001110001101 C38D × 1100001110010111 C397 ÷ 1100001110110111 C3B7
Page 107	A cipher is said to be *computationally secure* if breaking it requires computing power that is not available in practice.	A cipher is said to be *computationally secure* if any attack that can be carried out in realistic (efficient) time has only a negligible chance of breaking it.
Page 133	When $\omega_1 = \omega_2$...the Sbox is a ω_1-bit Sbox	When $\omega_1 = \omega_2$...the Sbox is an ω_1-bit Sbox
Page 139, RSA security	Nevertheless, post-quantum public key cryptosystems are being proposed (see e.g. [HPS98, BS08]) to protect communications after a quantum computer is built.	Nevertheless, post-quantum public key cryptosystems are being proposed (see e.g. [HPS98, BS08]) to protect communications after a sufficiently strong quantum computer is built.
Page 160, Example 3.2.4 last sentence	Then $\varphi_0(\boldsymbol{x}) = 0$.	Then $\varphi_0(\mathtt{0}) = 0$.
Page 168, third line	Let $n = pq$. We choose $e \in \mathbb{Z}^*_{\varphi(n)}$, and compute $d = e^{-1} \bmod \varphi(n)$.	Let $n = pq$. We choose $e \in \mathbb{Z}^*_{\varphi(n)}$, $e \neq 1$, and compute $d = e^{-1} \bmod \varphi(n)$.
Page 170, first paragraph	which is computationally infeasible according to property (c) of hash functions listed in Sect. 2.1.1.	which is computationally infeasible according to property (b) of hash functions listed in Sect. 2.1.1.
Page 177	$m = m_p y_q q + m_q y_p p \bmod n = 2 \times 2 \times 5 + 2 \times 2 \times 3 = 32 \bmod 15 = 2$.	$m = m_p y_q q + m_q y_p p \bmod n = 2 \times 2 \times 5 + 2 \times 2 \times 3 \bmod 15 = 32 \bmod 15 = 2$.
Page 209, last paragraph of Section 4.1.1	Similar to SPA, the attack does not require statistical analysis of the traces, only visual inspection is enough.	The sentence should be removed
Page 228	$\frac{\alpha}{2} = 1 - \Phi(z_\alpha) = 1 - \Phi(4.5) = \ldots$	$\frac{\alpha}{2} = 1 - \Phi(z_{\alpha/2}) = 1 - \Phi(4.5) = \ldots$
Page 236, Example 4.2.15	$\mathrm{E}\left[\mathrm{wt}\,(\boldsymbol{v})^2\right] = \frac{1}{\lvert\mathbb{F}_2^8\rvert} \sum_{\boldsymbol{v} \in \mathbb{F}_2^8} \mathrm{wt}\,(\boldsymbol{v}^2) = \ldots$	$\mathrm{E}\left[\mathrm{wt}\,(\boldsymbol{v})^2\right] = \frac{1}{\lvert\mathbb{F}_2^8\rvert} \sum_{\boldsymbol{v} \in \mathbb{F}_2^8} \mathrm{wt}\,(\boldsymbol{v})^2 = \ldots$
Page 248, Remark 4.3.1	For AES, the correlations between the first AddRoundKey outputs are higher than correlations between the first SubBytes operation outputs, that is why in...	For the PRESENT cipher, correlations among outputs from the initial `addRoundKey` operation are stronger than those between outputs of the initial `sBoxLayer`. Therefore, in...
Page 255, Step 8 last sentence	...when the target signal $\mathrm{wt}(\boldsymbol{v})$.	...when the target signal is $\mathrm{wt}(\boldsymbol{v})$.
Page 255, Step 10	...We argue that this is achievable ...is in procession of a clone device	...We argue that this is achievable ...is in possession of a clone device.
Page 262, last sentence	With our profiling traces, we can compute M_{signal} templates.	With our profiling traces, we can compute M_{signal} templates, with each template correspond to one possible value of the target signal.
Page 263, first paragraph	For our illustrations, when the target signal is $\boldsymbol{v}$, we will have 16 templates. And when the target signal is $\mathrm{wt}\,(\boldsymbol{v})$, we will have 5 templates.	For our illustrations, when the target signal is $\boldsymbol{v}$, we obtain 16 templates, each corresponding to a possible value of $\boldsymbol{v}$ from `0` to `F`. When the target signal is $\mathrm{wt}\,(\boldsymbol{v})$, we derive 5 templates, each corresponding to a Hamming weight value from 0 to 4.
Page 263 Template Step c	For a fixed key hypothesis $\hat{k}_i$, we divide the M_p attack traces from P-DPA Step 10 into M_{signal} sets, $A_1, A_2, \ldots, A_{M_{signal}}$, depending on the hypothetical target intermediate value $\hat{\boldsymbol{v}}_{ij}$ obtained in P-DPA Step 11. In particular, for an attack trace $\boldsymbol{\ell}_j$, let s_{ij} denote the index of the set that it belongs to. Namely $\boldsymbol{\ell}_j \in A_{s_{ij}}$ given key hypothesis $\hat{k}_i$. We are only interested in the leakages at the POIs for each attack trace $\boldsymbol{\ell}_j = (l_1^j, l_2^j, \ldots, l_q^j)$. Define $\boldsymbol{\ell}_{j,\mathrm{POI}} := (l_{t_1}^j, l_{t_2}^j, \ldots, l_{t_{q_{\mathrm{POI}}}}^j)$. With the mean vector $\boldsymbol{\mu}_{s_{ij}}$ and the covariance matrix $Q_{s_{ij}}$ obtained in Template Step b, we can compute the probability of $\boldsymbol{\ell}_j$ given $\hat{k}_i$ using the PDF ...	We are only interested in the leakages at the POIs for each attack trace $\boldsymbol{\ell}_j = (l_1^j, l_2^j, \ldots, l_q^j)$. Define $\boldsymbol{\ell}_{j,\mathrm{POI}} := (l_{t_1}^j, l_{t_2}^j, \ldots, l_{t_{q_{\mathrm{POI}}}}^j)$. For each key hypothesis $\hat{k}_i$ and attack trace $\boldsymbol{\ell}_j$, we compute the hypothetical target intermediate value given the knowledge of the associated plaintext. Let $\mu_{s_{ij}}$ and $Q_{s_{ij}}$ be the template for this hypothetical value, corresponding to $\hat{k}_i$ and $\boldsymbol{\ell}_j$, as obtained in Template Step b. The probability of $\boldsymbol{\ell}_j$ given $\hat{k}_i$ can then be computed using the PDF ...
Page 267	$\mu_1 = -0.039027, \quad \sigma_1^2 = 2.1679112 \times 10^{-6}$.	$\mu_1 = -0.039027, \quad \sigma_1^2 = 2.16437 \times 10^{-6}$.
Page 268, Figure 4.46 caption	The target signal is given by the exact value of the 1st Sbox output.	The target signal is given by the exact value of the 6th Sbox output.
Page 339, Figure 4.90 caption	Estimations of guessing entropy computed...	Estimations of success rate computed...
Page 334, below Equation (4.83)	The size of T1 is 8×4, and the storage required is $2^8 \times 2^4 = 2^{12}$ bits, or 2^9 bytes. The table T2 requires 16 bits of memory.	The size of T1 is 8×4, and the storage required is $2^8 \times 4 = 2^{10}$ bits, or 2^7 bytes. The table T2 requires $2^4 \times 4 = 64$ bits of memory.
Page 395, the last equation	$q = \gcd(s'^e - m, n) = \gcd(7^{11} - 2143) = \gcd(1977326741, 143)$.	$q = \gcd(s'^e - m, n) = \gcd(7^{11} - 2, 143) = \gcd(1977326741, 143)$.